

Quantum Networks

PAL

A team of researchers is actively testing an experimental quantum network based on quantum information exchange within the university campus environment. This network comprises several Quantum Processing Units (QPUs), between which direct quantum connections have been established for selected pairs of QPUs. It has been ensured that every pair of QPUs is connected, either directly or indirectly. In this context, two QPUs, A and B , are considered indirectly connected if there is a sequence of QPUs $A_1, \dots, A_k$ such that $k \geq 3$, $A = A_1$, $B = A_k$, and for all $i = 1, \dots, k - 1$, QPUs A_i, A_{i+1} are directly connected.

The topology of the considered network was proposed through careful calculation. The researchers initially determined the number of required QPUs and their placements. Subsequently, they created a scheme indicating which pairs of QPUs are suitable for establishing a stable direct connection. For each such pair, the cost of maintaining the corresponding connection was also determined, with this cost depending on various physical factors in the immediate vicinity. Ultimately, a solution with the minimal total maintenance cost was selected for implementation. Given that the costs of potential direct connections were pairwise different, the solution was unambiguously determined.

In the course of a new project, the researchers identified the need for a second network with similar properties to enhance the speed of experiment execution. Therefore, they are exploring suitable direct connections beyond the original plan, ensuring the interconnection of all QPUs into an independent communication network, without reusing or doubling direct connections from the first network. Their goal is once again to minimize the total maintenance cost for the second network.

However, they are also constrained by the fact that certain direct connections cannot be now utilized due to undesirable interference between the networks. Specifically, it is not possible to add any direct connection c that would create a cycle with more than D direct connections (including c) in the original network topology, where D is a constant derived by the researchers.

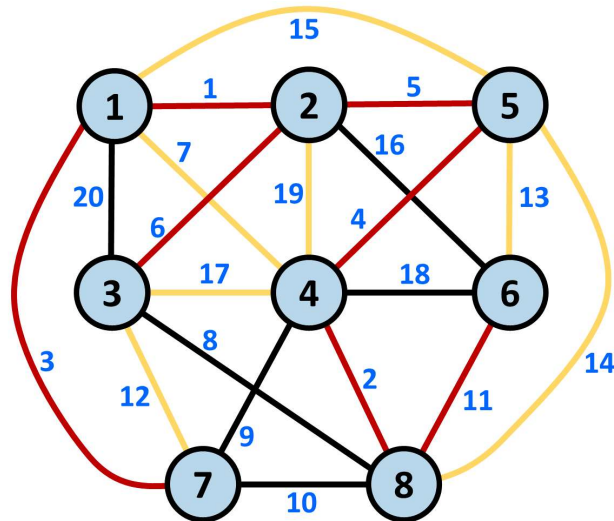


Figure 1. The diagram depicts QPUs as nodes, with potential direct quantum connections represented as edges. The blue numbers along these edges signify their maintenance costs. These costs are pairwise different. Connections of the first network are shown in red. Assuming $D = 4$, the second network is formed by the connections depicted in yellow. The total maintenance cost for the first and second network is 32 and 97, respectively. Both these costs are minimal possible. Importantly, the second network cannot include the connections with costs 8, 9, 10 and 16 since any of these connections would close a cycle with more than D edges (5 or 6) within the topology of the first network.

The task

We are given a list of QPU pairs for which it is possible to establish direct quantum connections, together with their maintenance costs. We also know the value of D . Our goal is to determine the total maintenance costs for both quantum networks created by the researchers.

Input

The first input line contains three integers Q , C , D , separated by spaces, where Q is the number of QPUs, C is the number of potential direct connections, and D is the constant established by the researchers. QPUs are numbered from 1 to Q . Next, C input lines representing potential direct connections follow. Each of them contains three integers Q_1 , Q_2 , M , separated by spaces, where Q_1 and Q_2 are numbers of QPUs to be connected and M is the maintenance cost of the connection. The connections are arranged in ascending order based on their unique costs.

It holds $Q \leq 5 \times 10^4$, $C \leq 6 \times 10^5$, $3 \leq D \leq 20$. The maintenance cost for each direct connection is not greater than 1.1×10^6 .

The input data guarantees the existence of a feasible solution.

Output

The output consists of two integers P_1 , P_2 separated by space. It holds $P_1 = M_1 \bmod 2^{16}$ and $P_2 = M_2 \bmod 2^{16}$, where M_1 and M_2 are total maintenance costs of the first and second network, respectively.

Example 1

Input

```
6 12 4
3 4 1
1 6 3
5 6 4
4 5 6
2 4 8
1 2 9
2 3 12
3 5 13
1 5 14
2 6 16
2 5 18
4 6 23
```

Output

```
22 78
```

Example 2

Input

```
8 20 4
1 2 1
4 8 2
1 7 3
4 5 4
2 5 5
2 3 6
1 4 7
3 8 8
4 7 9
7 8 10
6 8 11
3 7 12
5 6 13
5 8 14
1 5 15
2 6 16
3 4 17
4 6 18
2 4 19
1 3 20
```

Output

```
32 97
```

Example 3

Input

```
9 30 5
9 8 1
7 4 4
4 1 5
9 7 6
1 3 7
2 3 8
3 7 9
5 3 11
8 3 14
1 6 15
4 2 16
1 7 17
4 9 20
5 9 23
5 4 24
9 2 26
3 4 27
5 2 29
1 5 30
2 6 31
6 5 34
5 8 35
3 6 38
6 4 40
3 9 43
2 8 45
4 8 48
2 7 50
```

7 8 51
2 1 52

Output

57 192

The data of Example 2 is depicted in **Figure 1**.

Public data

The public data set is intended for easier debugging and approximate program correctness checking. The public data set is stored also in the upload system and each time a student submits a solution it is run on the public dataset and the program output to stdout and stderr is available to him/her.

[Link to public data set](#)